

Fischin' for Clearer Skies

Biomass-To-Jet Fuel: Using Gasification and Fischer-Tropsch Chemistry

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Transportation makes up

29%

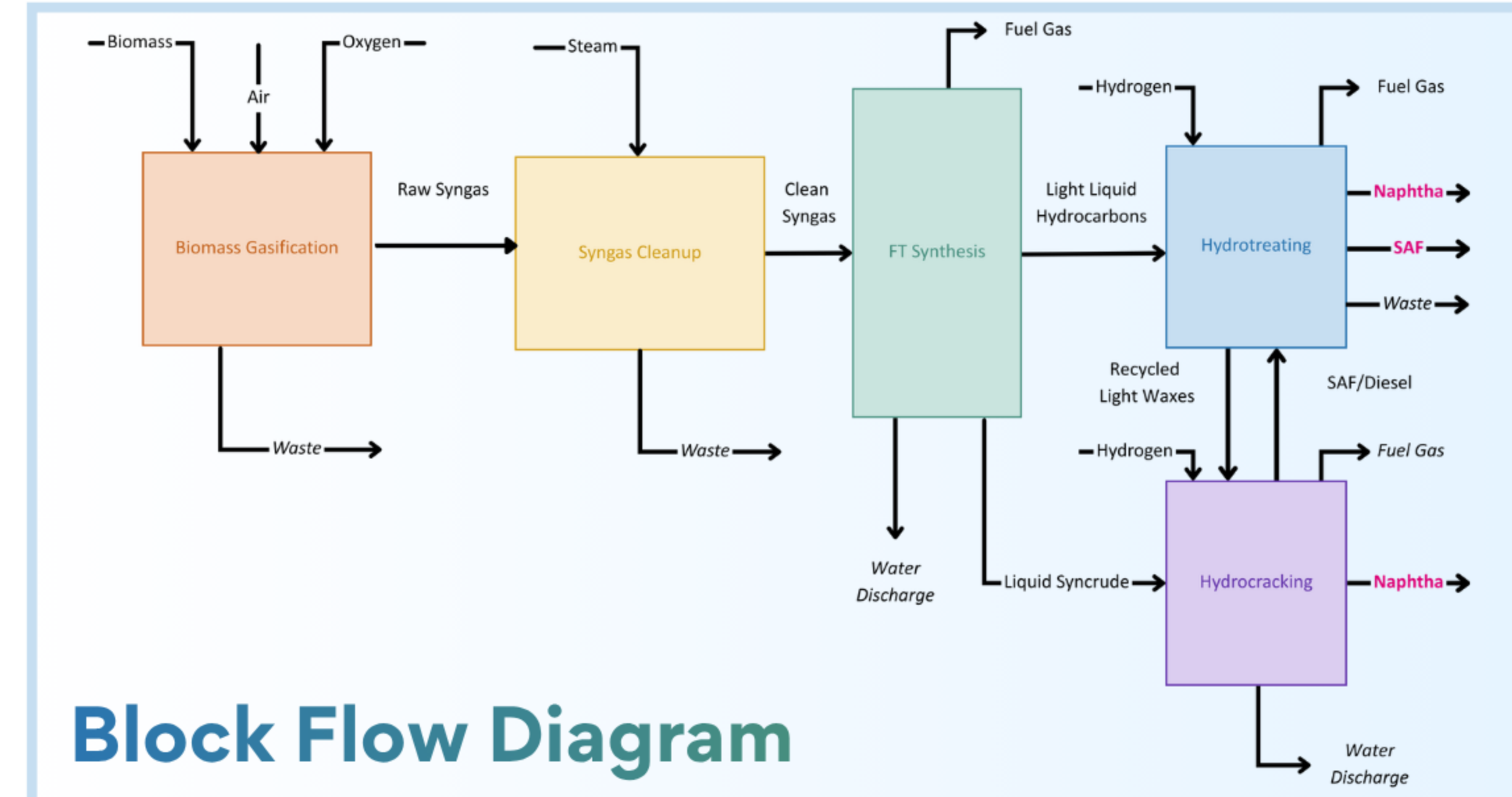
Of all greenhouse gas emissions ^[1]

Aviation travel makes up

9%

Of all transportation emissions

[1] U.S. Environmental Protection Agency. Sources of Greenhouse Gas Emissions: Transportation Sector. <https://www.epa.gov/ghgemissions/transportation-sector-emissions#sources>



Block Flow Diagram

Process Description

1

Woodchips are gasified to produce Syngas (CO+H₂)

2

Syngas is cleaned to produce correct ratio and purity

3

Clean Syngas is sent to Fischer-Tropsch Reactor to produce liquid hydrocarbons via catalytic reactions

4

Light Liquid Hydrocarbons are hydrotreated with hydrogen to produce final products

5

Liquid Syncrude and waxes are sent to be cracked down into useable SAF range hydrocarbons

Safety

Unit	Safety Concerns	Mitigation
Biomass Gasifier	High Temperatures	Temperature monitoring
	Flammable gasses	Gas leak detection
CO2 Absorber	CO2-rich solvent leaks	Secondary containment
	Column Flooding	Pressure monitoring
Fischer-Tropsch Reactor	Highly Exothermic Reaction	Temperature monitoring
	CO2 and H2 Leaks	Feed control, pressure relief
	Runaway temperatures	Temperature shutdown
Hydrotreater Vacuum Distillation Column	Vacuum Loss	See control loop
Hydrocracker	High pressure hydrogen	Pressure relief valve, flare
	Strong exothermic reaction	Temperature monitoring
	Hydrogen leak and fire risk	Gas leak detection

Design Basis

Saint James County, Louisiana
(U.S. Gulf Coast)

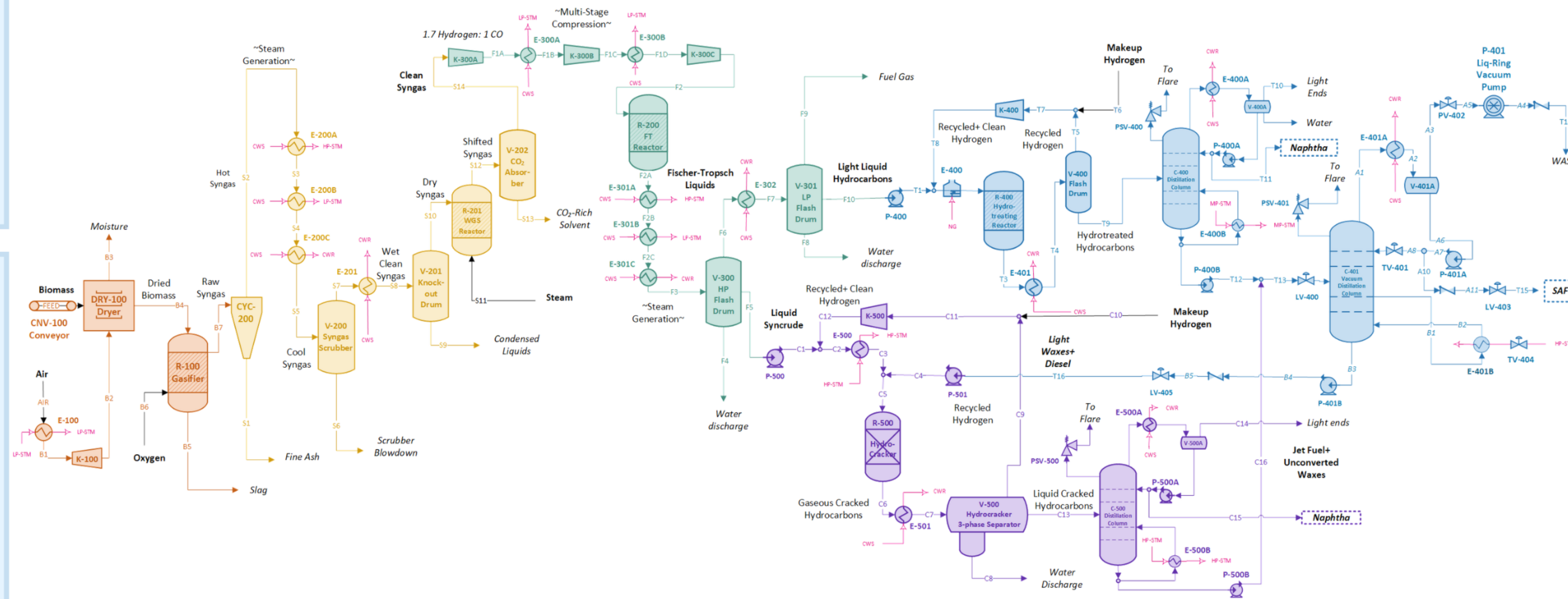
- Abundant forestry and agricultural biomass
- Strong hydrogen availability
- Deep-water port access
- High SAF demand from nearby major airports



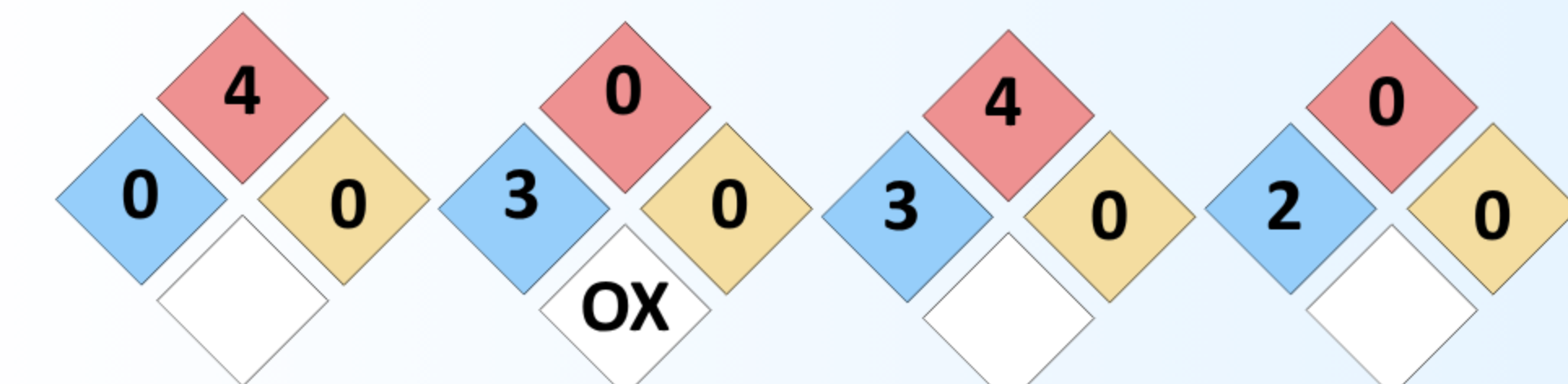
Plant Capacity

SAF	10,000 bbl/d
Naptha	3,500 bbl/d

Process Flow Diagram



Chemical Safety

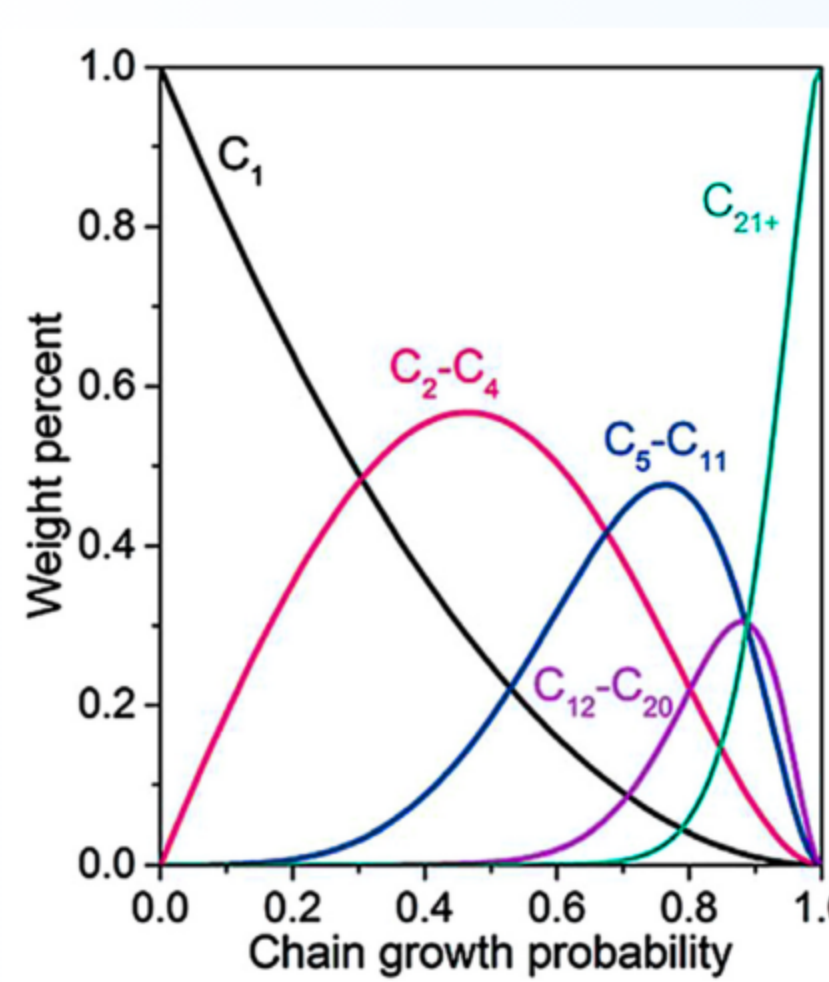


Hydrogen Oxygen Carbon Monoxide Carbon Dioxide

*Health Hazard *Fire Hazard *Instability Hazard *Special Hazard

Environmental Hazards

Type	Amount	Source
Air Emissions	470,000 tonne/yr	CO2, Light Ends
Wastewater	700,000 tonne/yr	Syngas Cleanup, Flash Drums & Separators
Solid Waste	10,000 tonne/yr	Ash



Chemistry

Fischer-Tropsch Reactions^[2]

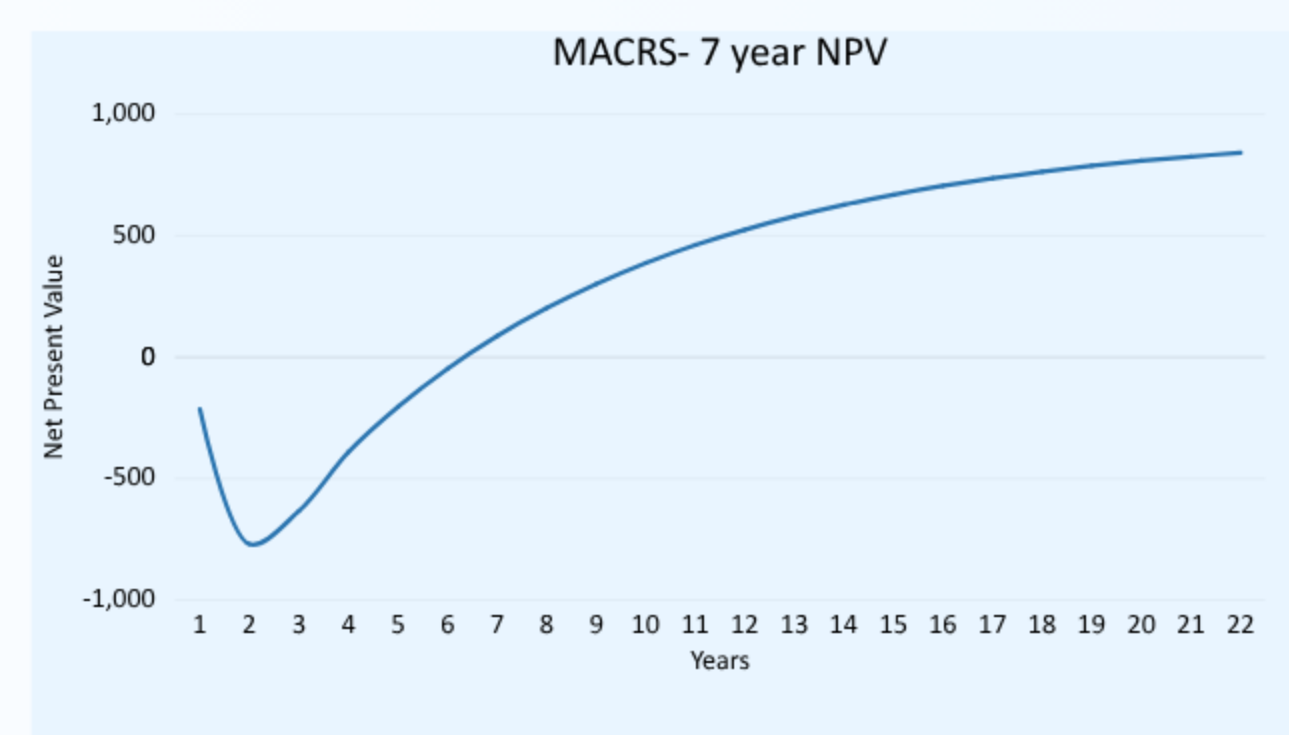
$n\text{CO} + (2n+1)\text{H}_2 \rightarrow \text{C}_n\text{H}_{(2n+2)} + n\text{H}_2\text{O}$
Molecular weight distribution according to Anderson-Schulz-Flory distribution
 $W_n/n = 1 - \alpha^2 \propto \alpha^{(n-1)}$

Water Gas Shift Reaction

$\text{CO} + \text{H}_2\text{O} \leftrightarrow \text{H}_2 + \text{CO}_2$
To maintain correct H₂:CO Ratio = 1.7:1

[2] Guo, L., Sun, J., Ge, Q., & Tsoubaki, N. (2018). Recent advances in direct catalytic hydrogenation of carbon dioxide to valuable C₂+ hydrocarbons. Journal of Materials Chemistry A, 6(44), 23244-23262.

Process Economics

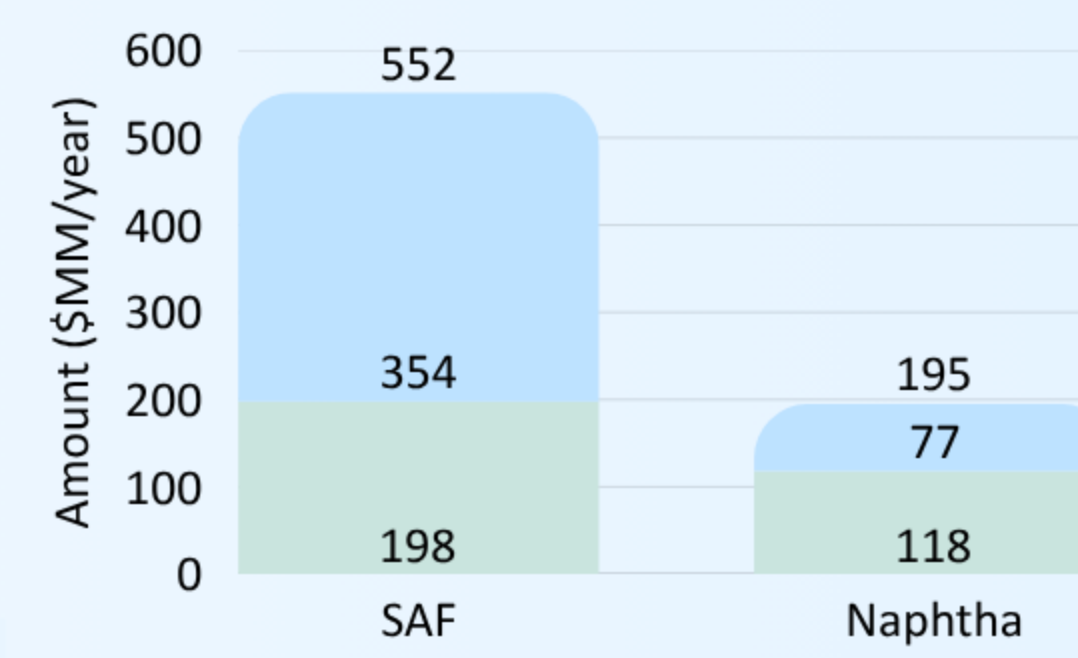


DCFOR (7-yr)	31.40%
Pay Back Period	2.14 years
Return on Investment	46.80%

How We Make Money

● Credit Based Revenue

● Revenue

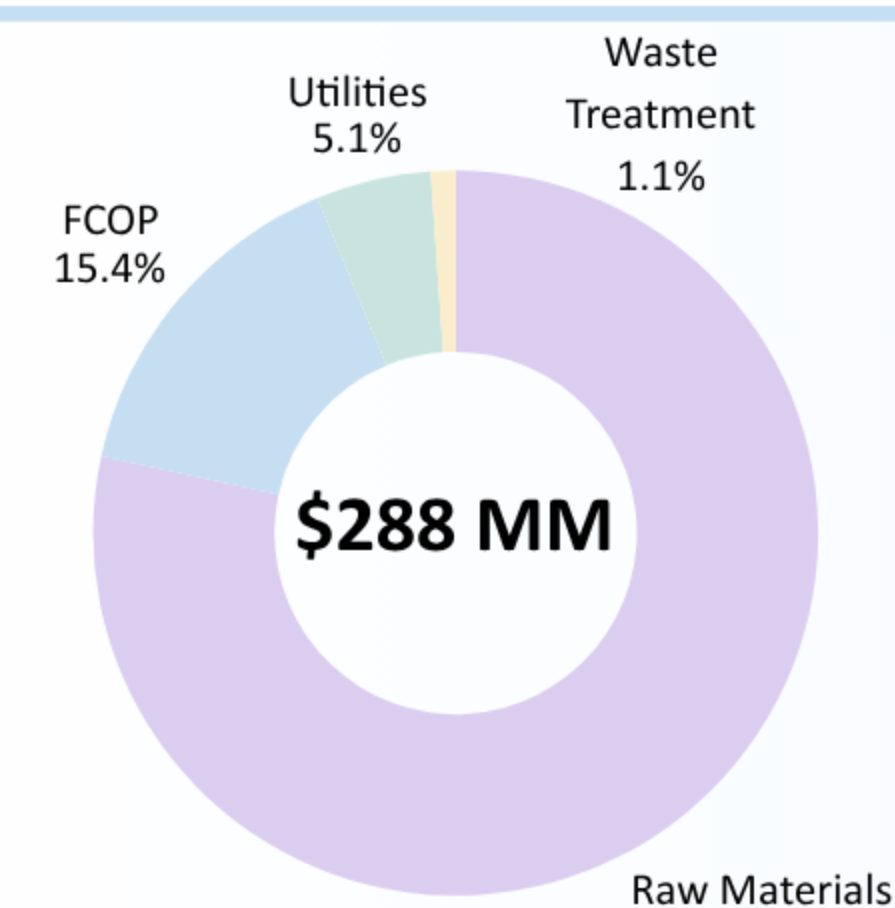
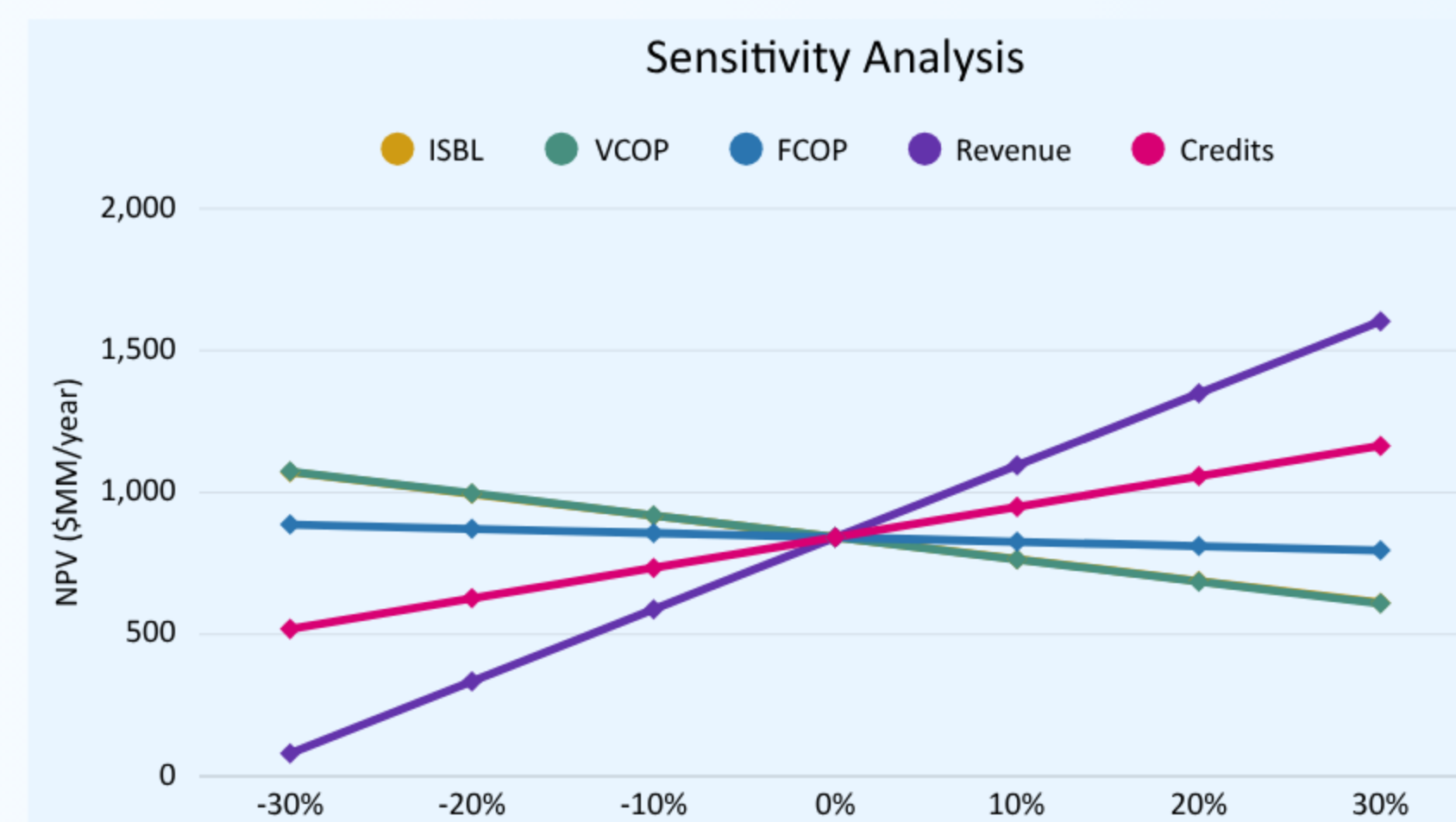


Income

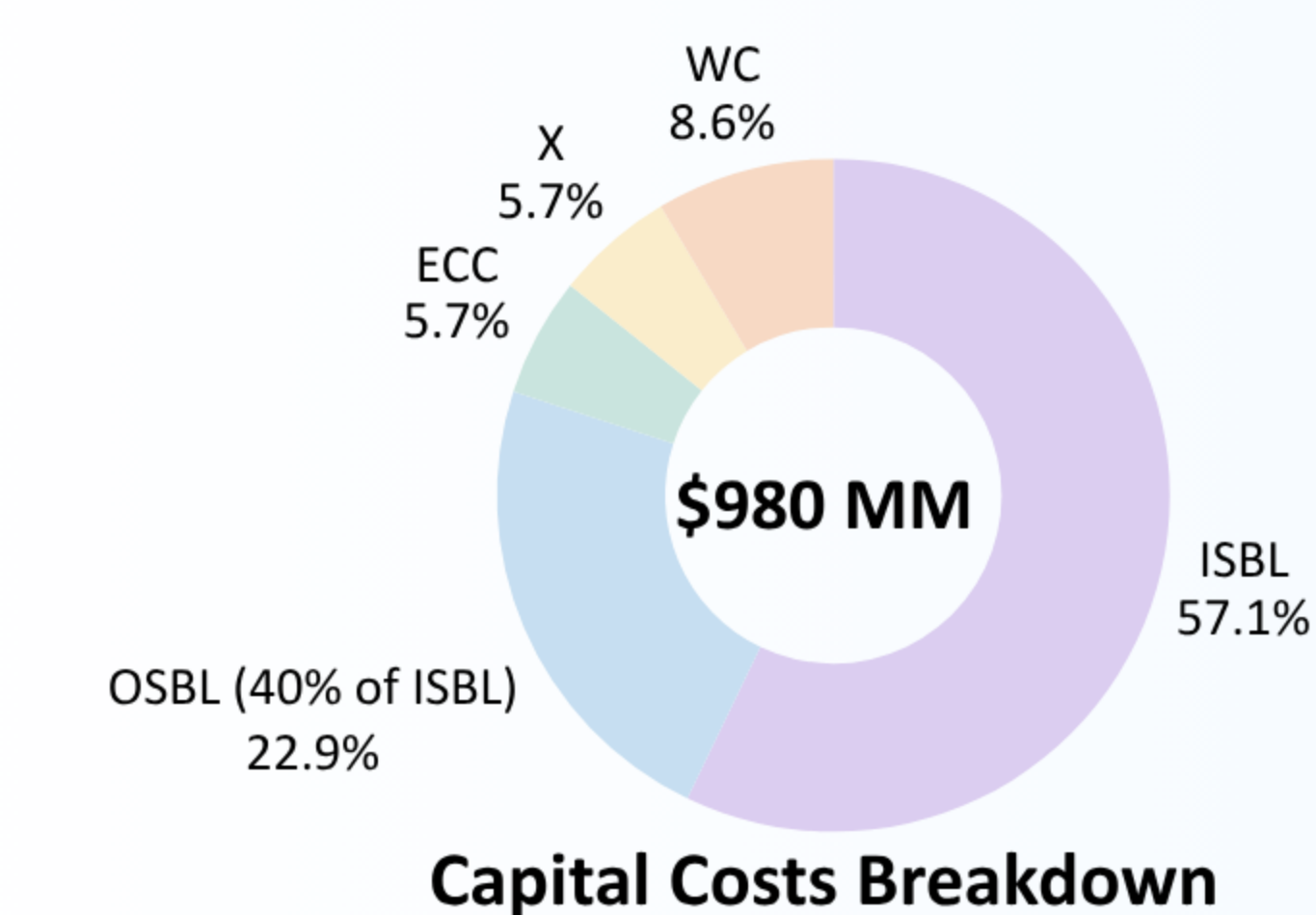
Total Revenue	750 \$MM/yr
Gross Profit	460 \$MM/yr

Constants

Tax Rate	26.5%
Interest Rate	15%

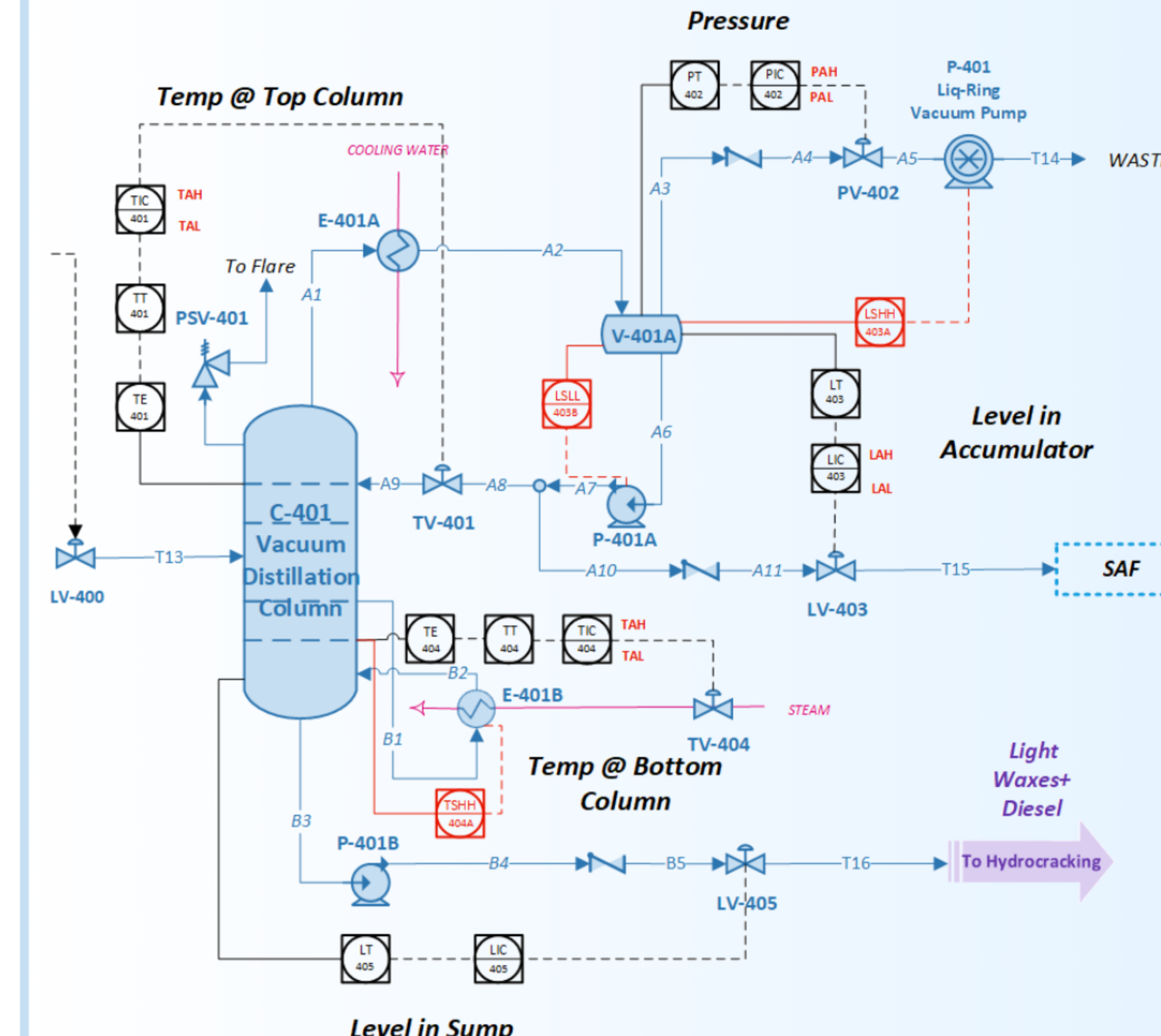


Cash Cost of Production



Capital Costs Breakdown

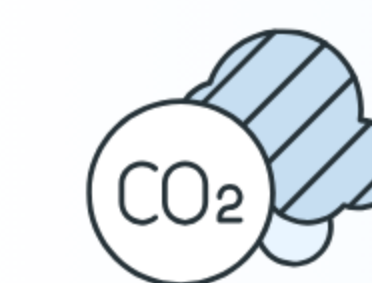
Controls for C-401



Final Recommendations



Reduce Carbon Footprint by: Electrolysis for Onsite Oxygen and Hydrogen Production



Improve plant efficiency: Heat Integration, Reduce feedstock, Catalyst Regeneration